

Simulation Study of Interactions Between Kinesin's Neck Linker and Motor Domain

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Abstract—The docking movement of kinesin's neck linker to the motor domain is the force generation step of kinesin. This docking movement is achieved by various non-bonding interactions between neck linker and motor domain. However, the relative strength and the structural basis for these interactions are still unclear. To elucidate these problems, we unbind the docked neck linker from motor domain using steered molecular dynamics and mutation simulations. We find that the surprisingly high strength of a structure called the “asparagine latch” arises from exquisite cooperation between hydrogen bonds and hydrophobic interactions. A β -sheet called the “cover-neck bundle” behaves like an elastic beam in its unbinding process. The hydrophobic interaction between ILE327 (amino acid sequence in 2KIN) and the hydrophobic pocket on the motor domain is strong and provides a fixed support for elastic bending of the cover-neck bundle.

Keywords—Kinesin, Neck linker, Molecular dynamics simulation, Non-bonding interaction.

INTRODUCTION

Conventional kinesin (kinesin-1)¹³ is a dimeric motor protein responsible for organelle transportation in cells.^{5,29,27} Through the nucleotide dependent binding and unbinding of neck linker (NL) to the motor domain,^{20,30} kinesin's two motor heads switch their positions on microtubule in a hand-over-hand manner.^{28,30} NL's docking to its binding site on the motor domain is thought to be the key force generation step,^{1,20,26,28,4}. The force generated during this process arises from the contributions of various interactions between NL and the motor domain. These

interactions mainly include the hydrophobic interaction between ILE327 (herein, we use amino acid numbering for 2KIN²⁴) and motor domain, the forming of a β -sheet which is called the “cover-neck bundle” (CNB) structure and the formation of the “asparagine latch” (ASN latch).^{8,9,11,12,15,22,23,30} These interactions are of quite different features and an understanding of the structural basis of each of these features will provide insight into kinesin's force generation mechanism.

Kinesin's NL is a short peptide (~ 14 amino acid residues) which connects the motor head and the coiled-coil stalk. NL is docked to the motor domain in the ATP-binding state and undocked from the motor domain in the ADP-binding or nucleotide-free state.²⁰ NL consists of three parts. The first part is the first three residues (LYS325, THR326 and ILE327) of NL, which form an extra turn at the C-terminal end of $\alpha 6$ when NL is in the docked state. The other two parts are $\beta 9$ and $\beta 10$. Correspondingly, docking of NL is accomplished in three steps, i.e., (1) extra turn formation, (2) $\beta 9$ docking and (3) $\beta 10$ docking to the motor domain. It has been revealed that docking of a highly conserved residue, ILE327, to a hydrophobic pocket on motor domain is a crucial step of NL's docking movement.^{30,12,22,23} Nitta *et al.*¹⁵ proposed that ILE327's docking to the hydrophobic pocket is the initial step of NL docking. Our investigation² also shows that the hydrophobic interaction between ILE327 and motor domain, together with the ATP binding induced rotation of the motor head, provides the driving force for the extra turn formation.

Hwang *et al.*^{8,9} studied the interactions between NL and motor domain by molecular dynamics (MD) simulation and found that the N-terminal strand of motor domain, $\beta 0$, contributes to the docking of $\beta 9$ by forming a CNB with NL. CNB structure is a β -sheet consisting of two strands and is biased to bend toward

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the motor domain due to twisting of the β -sheet. This special force-generating mechanism, i.e., forming CNB structure, is also shared by the kinesin-5 family member Eg5.^{3,4,19} In the work of Hwang *et al.*^{8,9} they found that ASN334 acts as a latch (ASN latch) that holds β 10 of NL firmly in the bound state. ASN334 locates at the beginning of β 10 and the latch function is mainly achieved by forming a backbone hydrogen bond (H-bond) between ASN334 and GLY77 on the motor domain. However, this ASN latch seems to have a surprisingly high strength that Hwang *et al.* need to use a strong force (> 400 pN) to break it within 1 ns. GLU336 of β 10 also forms backbone hydrogen bonds and even a salt bridge with LYS223 on motor domain. But, it is much easier to unbind GLU336 than ASN334.⁸ Therefore, the high strength of ASN latch still needs a reasonable explanation.

The NL's three parts dock to the motor domain in three different ways, each in which displays a special mechanical feature. To elucidate and compare these features on structural basis, we use steered molecular dynamics (SMD) and residue mutation method to simulate the unbinding process of NL under backward pulling force. Detailed analysis of the force-time curve obtained from constant velocity SMD (CV-SMD) simulation reveals the following results. First, the high strength of the ASN latch arises from the exquisite cooperation between H-bonds and hydrophobic interactions. Second, the CNB structure behaves like an elastic beam in NL's unbinding process. Third, the hydrophobic interaction between ILE327 and the hydrophobic pocket on the motor domain is strong and provides a fixed support for the elastic bending of CNB.

METHODS

The SMD simulations are performed by using NAMD (version 2.9)¹⁷ with force field CHARMM.¹⁸ The data analysis is performed with VMD (version 1.9.1).⁷ The non-bonded Coulomb and van der Waals interactions are calculated with a cutoff using a switching function starting at a distance of 13 Å and reaching zero at 15 Å. The structural model for simulation is based on the X-ray structure of rat kinesin, 2KIN.²⁴ It has docked NL and is in ATP-binding state. To save computational resources, we delete 2KIN's α 7 (the residues after ALA339), ADP and SO4. The motor domain is surrounded by explicit water and the spherical boundary condition is used. 2KIN does not have coordinates for the main part of L11 loop. We connect the two ends of L11 (VAL239 and ASN256) directly. Because our model does not include microtubule, we fix the C_α atoms of ASN264 and SER267 in

α 4 to mimic the microtubule-bound state. We also fix the C_α atoms of LYS315 and MET319 in α 6 to keep the relative position of α 6 and α 4. The water model is TIP3P.¹⁰ In all the simulations, the protein-water system is minimized for 30000 steps to remove bad contacts. Molecular drawings are produced by using Discovery studio 3.5 visualizer.

A pulling force is exerted on the NL's C-terminal end (C_α of ALA339) with a CV-SMD protocol to simulate the unbinding force arising from inter-head interaction. The direction of this force is toward microtubule's minus end, see Fig. 1a. The integration time step in the CV-SMD is 1 fs. The pulling speed of CV-SMD simulation is 0.1 Å/ps and trajectories are recorded every 1 ps. The force constant is 1 kcal/mol/Å². We perform two kinds of SMD simulations. In the first kind of simulations, the model is the normal structure of 2KIN. This simulation is referred to as CV-SMD-I. In the second kind of simulations, β 0 (residues before LYS10) is deleted from the normal structure of 2KIN. This simulation is referred to as CV-SMD-II.

RESULTS AND DISCUSSION

Features Of Interactions Between NL and Motor Domain

There are several non-bonding interactions between NL and motor domain, including H-bonds, salt bridges, hydrophobic interactions and water bridges⁶ (Fig. 1b). CV-SMD simulations are carried out to investigate the features of interactions (snapshots of the unbinding trajectory are shown in Fig. 2). In CV-SMD simulations, the time-dependent change of unbinding force reflects the relative strength of non-bonding interactions in the unbinding region of the protein. Before the non-bonding interaction in the unbinding region is broken, the unbinding force increases over time. After the non-bonding interaction is broken, the unbinding force decreases, as seen in Fig. 3a. According to the force-time curve in Fig. 3a, we divide the unbinding process into three stages, i.e., 0 ps to ~200 ps (the first stage), ~200 ps to ~510 ps (the second stage) and ~510 ps to ~760 ps (the third stage), respectively. The force-time curve after 760 ps corresponds to stretching of α 6 which cannot happen in the real situation and is ignored in our analysis.

The C-terminal end of NL has a residue, GLU336, which forms two H-bonds and a salt bridge with LYS223 of the motor domain. The destruction of these interactions contribute to the increase of unbinding force in the first stage (Fig. 3b). As seen from Fig. 3b, the unbinding force increases from ~10 ps to ~50 ps. In

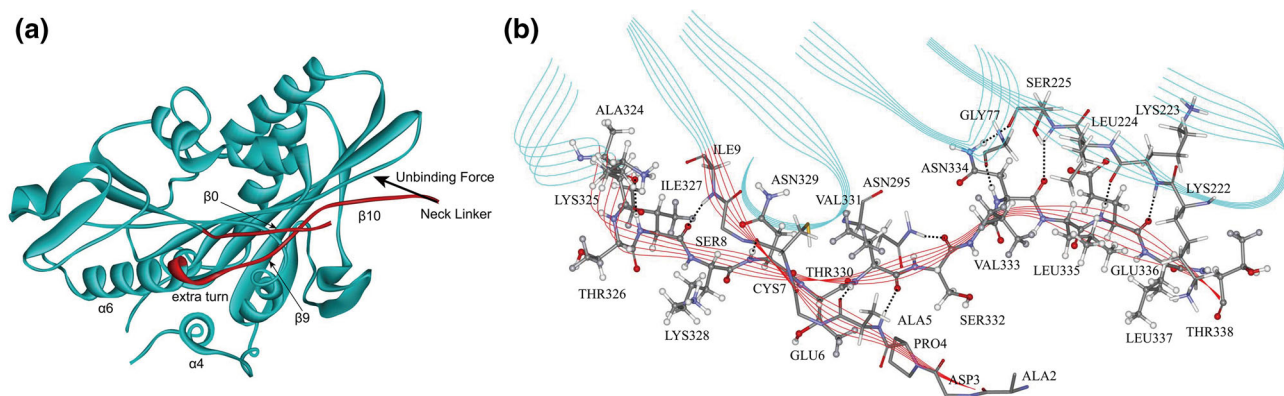


FIGURE 1. Neck linker and $\beta 0$ in the crystal structure of rat kinesin (2KIN). (a) The structure of kinesin's motor head. NL and $\beta 0$ are shown in red. The direction of the unbinding force in SMD simulation is indicated by the black arrow. (b) Conformation of NL (LYS325 ~ THR338) and $\beta 0$ (ALA2 ~ ILE9) in the docked state. Residues of NL are shown in ball and stick mode. The residues of motor domain and $\beta 0$ which have interactions with NL are shown in stick mode and the sidechain parts of $\beta 0$ are omitted. The hydrogen bonds between NL and motor domain and those between NL and $\beta 0$ are shown in dotted lines. NL consists of three parts: extra turn (LYS325, THR326 and ILE327), $\beta 9$ (LYS328 ~ SER332) and $\beta 10$ (VAL333 ~ THR338). The backbone hydrogen bond formed between ASN334 and GLY77 is the central hydrogen bond of ASN latch. The sidechain of ILE327 is totally buried in the hydrophobic pocket on the motor domain in the docked state of NL.

this period of time, the distance between charged groups of GLU336 and LYS223 decreases, indicating that the salt bridge contributes to the increase of unbinding force. In the same period of time, the backbone H-bonds between GLU336 and LYS223 are maintained, which also contribute to the increase of the unbinding force. After ~50 ps, the unbinding force decreases sharply. At the same time, the distance between charged groups of GLU336 and LYS223 increases, indicating that the decrease of unbinding force is mainly caused by weakening of the salt bridge. From ~70 ps to ~170 ps, the unbinding force increases. In this period of time, the backbone H-bonds between GLU336 and LYS223 are maintained, indicating that these two H-bonds contribute to the increase of unbinding force. At the same time, side chains of THR338 and GLU221 form a H-bond due to the conformational change of the unbinding process, which also contribute to the increase of the unbinding force. After the H-bond between THR338 and GLU221 is broken (indicated by the increase of the pink curve from 2 Å to 4 Å at ~170 ps), the unbinding force starts to decrease. After the two backbone H-bonds between GLU336 and LYS223 are totally broken, the unbinding force decreases until the end of the first stage. Under the action of backward unbinding force, the orientation of C-terminus of NL changes from pointing to the microtubule's plus end (forward) to pointing to the microtubule's minus end (backward). In this process, the unbinding force also drives the rotation of peptide planes of C-terminal residues, which also contributes to the increase of unbinding force (details not shown).

The second stage of force-time curve is a high peak as shown in Fig. 3a. According to the simulation trajectory, the increase of unbinding force (from ~200 ps to ~420 ps) reflects the unbinding process of ASN latch. In the simulations of Hwang *et al.*⁹ this ASN latch seems to have a surprisingly high strength that they need to use a strong force (> 400 pN) to break it within 1 ns. The appearance of the high peak of the force-time curve in this stage is consistent with the result of Hwang *et al.* In the crystal structure of 2KIN, there are two H-bonds among ASN334, GLY77 and SER225. ASN334 forms a backbone H-bond with GLY77 and a backbone H-bond with SER225. The structure after energy minimization in our simulation reveals another H-bond formed between the side-chain of ASN334 and the backbone of SER225 (Fig. 1b). It is interesting that the stabilities of these three H-bonds are very different. From Fig. 3c, the backbone H-bond between ASN334 and SER225 is not stable and is broken early in the simulation trajectory. The side-chain H-bond between ASN334 and SER225 and the backbone H-bond between ASN334 and GLY77 are stable. They are broken at ~420 ps and ~450 ps, respectively, that fit the position of the high peak of unbinding force in the second stage. In water environment, the effective strength of a H-bond is largely dependent on whether this H-bond is exposed to the surrounding water. From structural analysis, we find that these two H-bonds are protected by three hydrophobic residues (LEU335, VAL333 in NL and LEU224 in motor domain) which effectively cover the H-bonding region. The hydrophobic interactions between these hydrophobic residues also contribute to

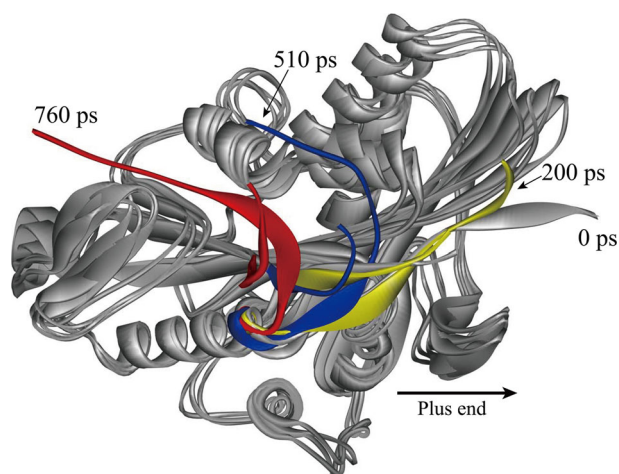


FIGURE 2. Snapshots of the NL unbinding trajectory in SMD simulation. Conformations of NL and $\beta 0$ at 200 ps, 510 ps and 760 ps are colored yellow, blue and red, respectively. They are end conformations of three stages of NL unbinding (see the text and Fig. 3a). At 510 ps, ASN latch is totally broken and $\beta 10$ is undocked. At 760 ps, the hydrophobic interaction between ILE327 and the motor domain is broken and $\beta 9$ is completely undocked. It is noted that NL and $\beta 0$ maintain CNB structure in the entire unbinding trajectory.

the increase of the unbinding force. It is important to observe that the surprisingly high strength of ASN latch and the correspondingly high peak of this stage arise from exquisite cooperation between H-bonds and hydrophobic interactions. An H-bond is a short-range interaction. The effective force range can be indicated by the distance between the donor hydrogen and acceptor oxygen within 1.75 to 2.5 Å.¹⁶ Comparatively, hydrophobic interaction is a medium-range interaction with effective force range within 7 Å between the two hydrophobic groups.¹⁴ In the ASN latch region, the amino acids participating hydrophobic interaction are leucine and valine, both of which have two methyl groups in their side chains. We thus use the distance between CB of valine and CG of leucine to indicate the hydrophobic interaction between side chains. When this distance is within 8 Å, the two hydrophobic side chains are in effective hydrophobic interaction. From Fig. 3c, the backbone H-bond between ASN334 and GLY77 and the two hydrophobic interactions maintain a long time until ~420 ps, at which the unbinding force reaches the top of the peak in this stage. The H-bond between side chain of ASN334 and backbone of SER225 is not as stable as the backbone H-bond between ASN334 and GLY77. These interactions in the ASN latch region are reasonably organized around the H-bond between ASN334 and GLY77 and we call this H-bond the central H-bond. On one side of the central H-bond, the hydrophobic residues effectively protect it from water attack. On the other side of the central H-bond, the H-bonding region between

ASN334 and SER225 effectively covers the H-bonding region of the central H-bond and protect the central H-bond from water attack from the other side (see Fig. 1b). Therefore, the central H-bond has a high stability like a H-bond in the interior of a protein. On the other hand, the stable central H-bond effectively limits the movement of the H-bonding and hydrophobic groups and effectively strengthens those other interactions of ASN latch. The exquisite cooperation between these interactions results in the high strength of ASN latch and the corresponding high peak of the force-time curve in the second stage. In the work by Rice *et al.*²¹, the V333A/N334A mutation effectively weakened the ASN latch and caused motility defects, which is consistent with our results and analysis here. After the central H-bond is broken, the unbinding force begins to decrease rapidly until the end of the second stage.

The third stage of the force-time curve reflects the unbinding process of CNB and the opening of extra turn. $\beta 9$ and $\beta 0$ form a CNB structure which has a tendency of bending to the motor domain,^{9,11}. However, $\beta 9$ has no direct strong interaction (e.g., backbone H-bond) with the motor domain. Therefore, the unbinding force of this stage is mainly to resist the bending of CNB structure. At the beginning of the third stage, the force-time curve fluctuates for about ~60 ps (from ~520 ps to ~580 ps), indicating that the bending of CNB is relatively weak in this period. From ~580 ps, the unbinding force increases approximately linearly (Fig. 3d). The linear increase of the unbinding force reflects that the resistant force from CNB bending is proportional to the displacement of the end of CNB, i.e., CNB behaves like an elastic beam. In this simulation, the stiffness coefficient of CNB is ~31 pN/Å (estimate by linear fitting method). The bending of an elastic beam needs a fixed support which, in this case, is achieved by the hydrophobic interaction between ILE327 and hydrophobic pocket on the motor domain. ILE327 is highly conserved in the kinesin superfamily. Hydrophobic interaction between ILE327 and hydrophobic pocket is strong because the change of the solvent accessible area in this hydrophobic interaction is just the entire surface area of the hydrophobic side chain of ILE327.^{14,25} When the hydrophobic interaction between ILE327 and hydrophobic pocket is broken, i.e., ILE327 is pulled out of the hydrophobic pocket, the unbinding force begins to decrease (at ~750 ps). After CNB structure is sufficiently stretched, the unbinding force starts to increase again since then the resistant force arises from the stretching of $\alpha 6$.

The above simulation results show that the surprisingly high strength of ASN latch arises from exquisite cooperation between H-bonds and hydro-

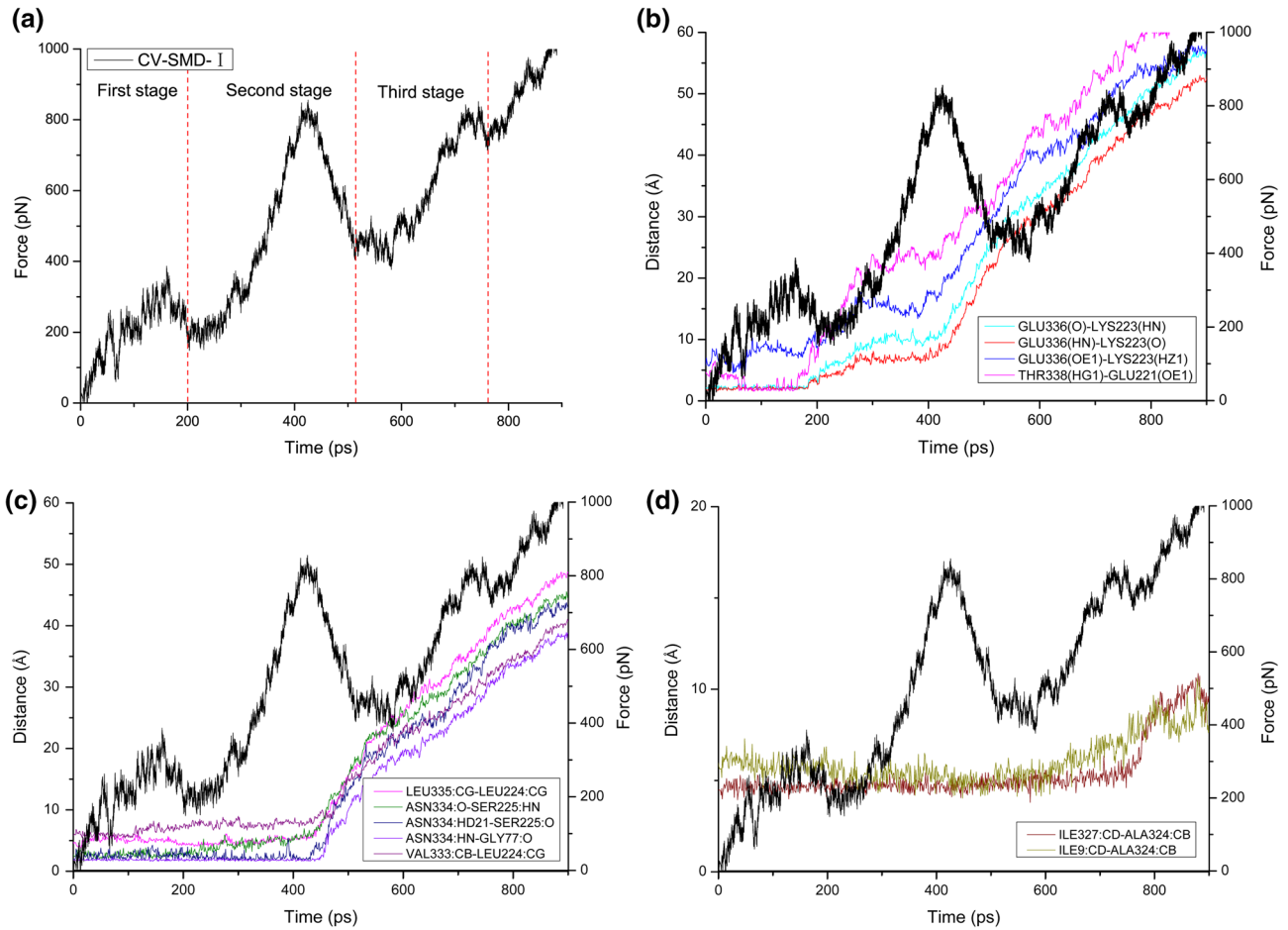


FIGURE 3. Force-time curve and distance-time curves of the unbinding simulation. (a) Force-time curve of the unbinding simulation. Three stages are separated by dotted red line. The first two stages reflect the unbinding process of $\beta 10$. The third stage reflects the unbinding of $\beta 9$ (and CNB) and the opening of extra turn. (b–d) Matching of the distance-time curves and the force-time curve for the first, second and third stage of the unbinding simulation, respectively. The black curve is the force-time curve and the colored curves are distance-time curves for the representative atoms of the non-bonding interactions between NL and the motor domain. An abrupt increase of the distance-time curve indicates breaking of the corresponding interaction. There is a special group of interactions corresponding to each stage in the force-time curve. Before the breaking of the interactions, the force-time curve goes up; after the breaking of the interactions, the force-time curve goes down. Matching the two kinds of curves clearly shows which interactions are responsible for the changes of the force-time curve. The hydrophobic interaction between two hydrophobic residues is indicated by the distance between the carbon atoms of the methyl groups of the two residues.

phobic interactions. The CNB structure behaves like an elastic beam in its unbinding process. The hydrophobic interaction between ILE327 and the hydrophobic pocket on the motor domain is strong and provides a fixed support for elastic bending of CNB structure.

In our simulations, CNB structure is maintained in all the simulation time of the NL unbinding process. This is a very interesting result because $\beta 0$ and NL are exposed to the surrounding water during the whole pulling process. Through analyzing the structure and the conformational change of amino acids on NL and $\beta 0$ in the simulation trajectory, we find that the side chains of residues on $\beta 9$ and $\beta 0$ can effectively protect the backbone H-bonds of CNB from water attack. In

addition, from the geometrical feature of the CNB structure in the docked state, it is a natural result that the CNB structure is maintained during the unbinding process under a backward force. We thus propose that CNB structure should exist in the initial conformation of NL docking and is maintained in the whole walking cycle.

CV-SMD SIMULATIONS WITHOUT $\beta 0$

In the simulation of CV-SMD-II, we delete $\beta 0$ and study the unbinding process in the same condition as that of CV-SMD-I. The force-time curve in this simulation also has three stages similar to that of

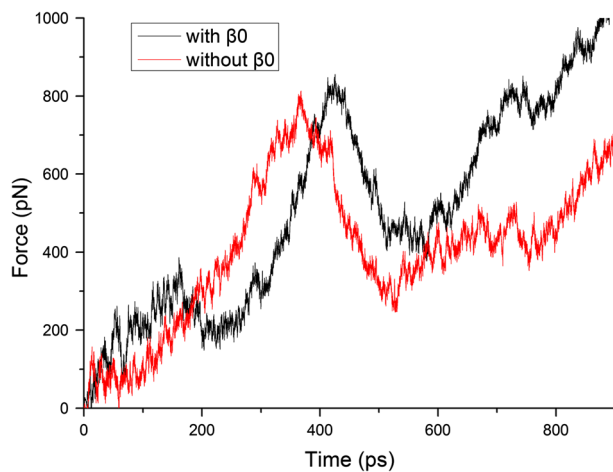


FIGURE 4. Comparison between the force-time curves of CV-SMD-I (with $\beta 0$) and CV-SMD-II (without $\beta 0$) simulations. The mutant model of CV-SMD-II has no $\beta 0$, i.e., it has no CNB structure. Thus, the curve in the third stage of CV-SMD-II is much lower than that of CV-SMD-I, whereas the two curves have similar structures in the second stage, indicating that the ASN latch mechanism is the same for the models in the two simulations.

CV-SMD-I (Fig. 4). As seen from Fig. 4, the stage one and stage two of the force-time curve in both CV-SMD-I and CV-SMD-II are similar. These two stages of the force-time curve reflect the unbinding process of $\beta 10$ which is not influenced by the deletion of $\beta 0$. However, in the third stage of force-time curve in CV-SMD-II, the unbinding force is weaker than that of CV-SMD-I (Fig. 4). Therefore, the CNB structure has strong tendency to dock to the motor domain, which is consistent with the results of Hwang *et al.*⁹

Note here that a recent literature⁴ has studied the CNB structure and its function in both kinesin-1 and kinesin-5 families by experiments and also molecular dynamics simulations. It has been found that the CNB structure is a force generation element and the NL of kinesin-5 makes additional contacts with the core of the motor head *via* loop L13. In kinesin-1, however, the NL interacts weakly with L13 due to different amino acids sequences between kinesin-1 and kinesin-5. Our simulation results about the NL's CNB region are consistent with Ref. 4. Considering the structural similarities between kinesin-1 and kinesin-5, the hydrophobic interaction at the root of NL and the interactions between $\beta 10$ and the motor head discussed in this paper may also exist in kinesin-5, which deserves to be further studied.

CONCLUSIONS

The force-time curve obtained from CV-SMD simulation can effectively reflect the mechanical features of

the unbinding region of a protein. We investigate the unbinding process of kinesin's NL using CV-SMD method. We find that the high strength of ASN latch arises from exquisite cooperation between H-bonds and hydrophobic interactions. The CNB structure has a strong docking tendency and behaves like an elastic beam in its unbinding process. The hydrophobic interaction between ILE327 and the hydrophobic pocket on the motor domain is strong and provides a fixed support for the elastic bending of CNB. These findings provide structure-based insights into the force generation mechanism of kinesin's NL.

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